Oss. Fissate le basi $\mathcal{B}$, $\mathcal{F}$, la corrispondenza

$f \mapsto M_{\mathcal{B}\mathcal{F}}(f)$ è una corrispondenza biunivoca

$\mathrm{Hom}(V,W) \to \mathrm{Mat}_{m\times n}(\mathbb{R})$

$\forall v \in V$

$M_{\mathcal{B}\mathcal{F}}(f)\ C_{\mathcal{B}}(v)$

$\|$

$C_{\mathcal{F}}(f(v))$

Esercizio

$\mathbb{R}[x]_{\le 2} \xrightarrow{D} \mathbb{R}[x]_{\le 1}$ (la derivata)

$\mathcal{B}$ $1, x, x^2$ $\quad$ $\mathcal{F}$ $1+x, 1-x$

$\forall \lambda_1, \lambda_2 \in \mathbb{R}$ $\forall p_1, p_2 \in \mathbb{R}[x]_{\le 2}$

$(\lambda_1 p_1 + \lambda_2 p_2)' \stackrel{?}{=} \lambda_1 p_1' + \lambda_2 p_2'$

$\forall p \in \mathbb{R}[x]_{\le 1}$

$a(1+x) + b(1-x) = p(x) = c_1 x + c_0$

$(a+b) + (a-b)x = c_1 x + c_0$

$\begin{cases} a+b = c_0 \\ a-b = c_1 \end{cases}$ $\quad \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$